

Erik Winston

Arlington, VA, 22203 | winstoe@rose-hulman.edu | 703 – 589 – 6784

Education:

Master of Mechanical Engineering, Bachelor of Science, Mechanical Engineering (*Combined BS/MS*) **May 2027**
Rose-Hulman Institute of Technology – Terre Haute, IN **GPA: 3.50**
Related Courses: Thermodynamics, Control Systems, Machine Component Design, Manufacturing, FEA, Propulsion Systems, Measurement Systems, Numerical Methods, Dynamic Systems, Mechanical Systems, Fluid Dynamics, Heat Transfer, Statics, Mechatronics, Technical Communications

Skills:

Technical: Geometric dimensioning and tolerancing, Manual mill and lathe, Computer numerical control mill, lathe and waterjet, 3D printing
Software: MATLAB, SOLIDWORKS, ANSYS, Autodesk Fusion 360, Autodesk AutoCAD, Autodesk Inventor, Arduino Integrated Development Environment, SkyCAD, C++, Productivity Suite (PLC Ladder Logic), FlutterFlow, Python
Language: Fluent Spanish (written and spoken)

Work Experience:

General Atomics EMS – Mechanical Intern **June 2025 – August 2025, June 2026 – Present**

- Invented a novel satellite attitude control system testing method utilizing an ultra-low friction ferrofluid enclosure and magnetic platform to evaluate tipoff and consistency of satellite deployment process
- Created 4 experimental piezo valve designs in a satellite attitude control system to improve characterization for control algorithm
- Developed and analyzed a cup-cone interface between a satellite and deployer to maximize safe preload and eliminate 60-minute shimming process during assembly
- Analyzed dynamics of a center of gravity and moment of inertia test setup, wrote procedure and performed a dry run on a satellite mass simulator using a FARO laser tracker
- Prototyped and fabricated a testing rig for a chip resistor burn wire mechanism to release a satellite solar array
- Investigated and evaluated methods for precise velocity reduction of a satellite during deployment testing using angled ball bearings and frictionless surfaces for client demonstration
- Performed vibration testing and post-processing data analysis for satellite flight units
- Assembled satellite flight units, attitude control system valves, and vehicle deployer

Project Experience:

Rose Propulsion Laboratory (Rose Rocketry) **September 2023 – Present**

- **Controls Lead** – Led 6-person team and managed top-level design for the control algorithm and instrumentation of a liquid rocket engine
- **Test Director** - Organized and officiated the hotfire, coldflow, hydrostatic, and igniter tests for the Liquid Engine 1 test campaign
- **Operations Lead** – Implemented an administrative restructuring and a task tracking system, prepared onboarding presentations and resources for new members, restructured file system and established workflows
- Designed the wiring diagram for a PLC to interface with 30 pressure, temperature, and limit switch sensors, actuate pneumatic valves, and manage communications and commands to the engine feed system
- Created a PLC ladder logic program to communicate with ground station equipment to monitor feed system conditions and perform engine testing procedures and safety shutdowns
- Developed a control algorithm and model for a liquid rocket lander to test different control strategies and enable controller co-design
- Constructed an electrical panel enclosure to house controls hardware, routed and secured sensor and actuator wiring

Milwaukee Electric Tool Corporation Pipe Camera Skid **February 2024 – May 2024**

- Designed and fabricated a camera skid to provide better maneuverability and increased visibility to the Milwaukee Pipe Camera
- Practiced iterative design and testing following an engineering process model
- Communicated and presented the final product to Milwaukee Tool in a formal report

University Student Launch Initiative (Rose Rocketry) **November 2024 – May 2025**

- Designed payload for rocketry mission, programming a STM32 to collect pressure, altitude, temperature, and other flight data from 8,000 ft, then transmit data to a ground station using APRS protocol

Variable-Shape Sailboat Keel **Sep 2022 – May 2023**

- Designed, modeled in Fusion 360, and 3D printed rapid prototypes for a novel sailing keel to direct hydrodynamic lift
- Fabricated final design and tested in a continuous pool to evaluate performance
- Produced and presented an IEEE report detailing research and results

Leadership Experience:

Rose Rocketry Executive Board **May 2025 – May 2026**

- **Secretary** – Managed social media, community outreach, and recruitment efforts

Stellantis EcoCAR Innovation Challenge **May 2026 – Present**

- **Vehicle Systems Engineering Technical Lead** – Selected to direct the systems engineering side of the EcoCAR challenge